MOGO: Resolving the Connector Compliance Paradox in Modular Soft Robotics

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Abstract—Modular soft robotics promises rapid prototyping through reconfigurable building blocks, yet existing systems suffer from the "Connector Compliance Paradox": rigid connectors (CCR > 100) create stress concentrations at soft-rigid interfaces. We present MOGO, a ball-and-socket connector system achieving CCR ≈ 1 through single-material TPU construction. Our key insight is that connector compliance enhances rather than degrades robustness—soft sockets conform to balls under load, increasing contact area and creating vacuum-assisted retention. Through iterative prototyping from SLA interlocking ribs to FDM ball-and-socket joints, we developed optimized infill strategies: 20–30% rectilinear sockets for easy insertion (25 N) paired with 50% gyroid balls for high retention (55 N), yielding a force ratio of 2.2. FEA validated failure modes under torsion (45° unlock), bending, and tension. The system demonstrates >500-cycle durability with 2-second assembly time, establishing that fully soft modular connectors are viable for reconfigurable soft robotic assemblies.

Index Terms—soft robotics, modular robots, compliant connectors, ball-and-socket joint, TPU, 3D printing

I. Introduction

Soft robotics has demonstrated transformative capabilities in human-robot interaction, adaptive manipulation, and morphological adaptability that rigid robotic systems cannot achieve [1]. These inherently compliant systems offer unique advantages including safe human interaction, conformable grasping, and continuous deformation. However, despite significant advances in actuation, sensing, and fabrication methods, soft robotics remains largely confined to specialized research environments. Each soft robotic application currently requires custom engineering from conception through fabrication, creating substantial barriers for researchers, educators, and makers.

The central vision motivating this work is simple yet ambitious: what if building soft robots was as intuitive as assembling LEGO bricks? The remarkable success of LEGO as a construction platform stems from connections that are simultaneously easy to establish and mechanically robust—a combination that has remained elusive in soft robotics.

A. The Connector Compliance Paradox

Our comprehensive literature review revealed a consistent pattern across existing modular soft robotic systems: all rely on rigid or semi-rigid connectors as interfaces between soft components. This creates what we term the "Connector Compliance Paradox"—the mechanical discontinuity between compliant modules and rigid connectors that fundamentally limits system performance.

To quantify this paradox, we introduce the Connector Compliance Ratio (CCR):

$$\mathrm{CCR} = \frac{E_{\mathrm{connector}}}{E_{\mathrm{module}}} \tag{1}$$

where $E_{\mathrm{connector}}$ is the elastic modulus of the connector material and E_{module} is the elastic modulus of the soft module material.

Analysis of three representative approaches from the literature reveals the magnitude of this problem (Table I). All existing systems exhibit CCR $\gg 1$, with values ranging from 100 to 5000, indicating connectors that are two to four orders of magnitude stiffer than the soft components they connect.

TABLE I
Connector Compliance Ratio Analysis of Existing Systems

System	Module	Connector	CCR
Soft LEGO [2]	Tango (1-10 MPa)	ABS (2000 MPa)	200–2000
Soft Origami [3]	E630/E650 elastomer	Rigid plastic	>100
Interlock Print [4]	Agilus30 (0.3 MPa)	3D plastic	∼5000
MOGO	TPU (20 MPa)	TPU (20 MPa)	1–10

This stiffness mismatch creates multiple interrelated problems: stress concentrations at material boundaries, compliance discontinuities limiting overall system deformation, and failure-prone rigid-soft interfaces under repeated loading.

B. Research Question and Contribution

This work addresses: Can a single-piece, all-soft connector maintain high mechanical strength across multi-axis loading while remaining user-assemblable?

Our principal contribution is demonstrating that the Connector Compliance Paradox can be resolved through uniform-material ball-and-socket geometry, with the surprising finding that compliance enhances connection robustness through increased contact area, stress distribu-

tion, misalignment tolerance, and vacuum-assisted retention.

II. Materials and Methods

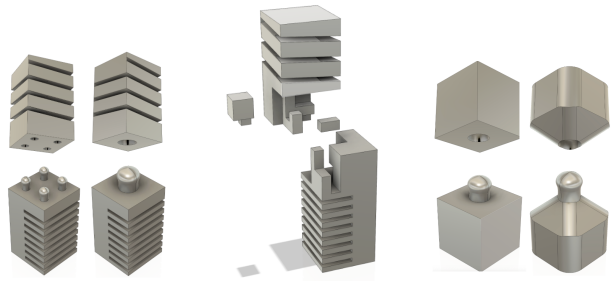
A. Design Evolution Through Three Manufacturing Phases

The MOGO connector system emerged through three distinct manufacturing phases (Fig. 1), with each iteration informing critical design decisions.

Phase 1 (SLA Interlocking Ribs): Initial prototypes drew inspiration from computationally designed mechanical interlocks [4], employing multi-part assemblies with interlocking ribbed structures. Fabrication utilized a Formlabs SLA 3D printer with Flexible Resin 80A, followed by IPA wash and UV curing. While parts were flexible, they proved difficult to attach reliably, and the multi-part assembly was impractical for ease-of-use goals.

Phase 2 (SLA Simplified Structure): The second phase aimed to simplify while trying different parameters. This iteration explored LEGO-style studs of various diameters combined with ribbed interfaces, along with central through-holes for potential cable routing. Testing revealed that while parts remained flexible, they still required precise orientation during assembly, limiting accessibility to novice users.

Phase 3 (FDM Ball-and-Socket): Insights from Phases 1 and 2 motivated a fundamental change in both fabrication method and connection geometry. We pivoted to FDM printing with thermoplastic polyurethane (TPU) using a ball-and-socket mechanism. Fabrication utilized an Anycubic Kobra 3 FDM printer with TPU filament. The ball-and-socket geometry was selected for several advantageous properties: passive geometric guidance facilitating easier connection, spherical contact interface enabling distributed load transfer, angular compliance accommodating misalignments, and single-piece fabrication with no post-print assembly.



(a) Phase 1: SLA ribs (b) Phase 2: Simplified (c) Phase 3: Ball-socket

Fig. 1. Manufacturing evolution. (a) Complex interlocking ribs in Flexible Resin 80A—too many parts. (b) Simplified ribbed modules with studs—difficult to attach. (c) Final ball-and-socket in TPU—simple and robust.

B. Infill Density and Pattern Optimization

A systematic study was conducted to optimize internal structure and mechanical performance. The key insight was that socket and ball modules have fundamentally different functional requirements necessitating distinct infill strategies.

Socket modules must deform sufficiently to allow ball insertion while providing adequate elastic restoring force for retention. Lower stiffness at the socket rim facilitates insertion. Ball modules must maintain shape under compression during insertion and resist deformation under pull-out loads, favoring higher stiffness.

Based on these requirements, socket modules were fabricated at 20%, 30%, and 40% infill density using Rectilinear and Grid patterns. Ball modules were produced at 30%, 40%, and 50% infill density using the Gyroid pattern, selected for its isotropic mechanical properties. Three specimens were fabricated for each condition, with each pair subjected to five insertion/removal cycles.

C. Finite Element Analysis

Structural validation employed SOLIDWORKS Simulation with nonlinear static analysis. The hyperelastic material model for TPU utilized Mooney-Rivlin parameters: $C_{10} = 0.69$ MPa, $C_{01} = 0.173$ MPa, density = 1.12 g/cm^3. The mesh consisted of second-order 3D tetrahedral elements (24,847 nodes, 15,234 elements) with curvature-based refinement at contact surfaces. Contact friction coefficient $\mu = 0.4$ based on TPU-to-TPU sliding.

Three loading scenarios were evaluated: torsion (4–5 N·m, based on NASA anthropometric data for hand strength [5]), bending (0.5–2.54 N, based on ergonomics research [6]), and tension for pull-out resistance.

D. Force Testing Protocol

Physical testing employed a Mark-10 Series 5 digital force gauge mounted on a motorized Mark-10 test stand. The test fixture secured the base module to the stand platform while the force gauge attached to the mobile module. The stand applied displacement at approximately 10 mm/min with the gauge in peak force capture mode. Two measurements were recorded: insertion force (user effort for assembly) and pull-out force (connection retention strength).

III. Results

A. Force Testing Results

Quantitative force measurements revealed strong dependencies on infill parameters enabling systematic optimization (Tables II, III).

Lower infill density and rectilinear pattern minimize insertion force by increasing socket compliance. The rectilinear pattern's straight-line structure allows easier radial expansion compared to the grid pattern's interconnected cells.

TABLE II
Socket Insertion Force vs. Infill Configuration

Infill	Rectilinear (N)	Grid (N)
20%	25.3 ± 2.1	28.7 ± 2.4
30%	38.2 ± 3.0	42.5 ± 2.8
40%	52.6 ± 3.7	56.1 ± 3.2

TABLE III
Ball Pull-out Force vs. Infill (Gyroid Pattern)

Infill Density	Pull-out Force (N)
30%	35.8 ± 2.9
40%	47.3 ± 3.5
50%	55.2 ± 4.1

The gyroid pattern's triply periodic minimal surface structure distributes stress evenly, preventing localized collapse. Higher density increases effective stiffness, maintaining ball shape under pull-out loads.

Optimal Configuration: 20–30% rectilinear sockets paired with 50% gyroid balls yield force ratio $\eta = 55.2/25.3 = 2.18$—significantly more force required to separate than to connect, a desirable characteristic for stable assemblies.

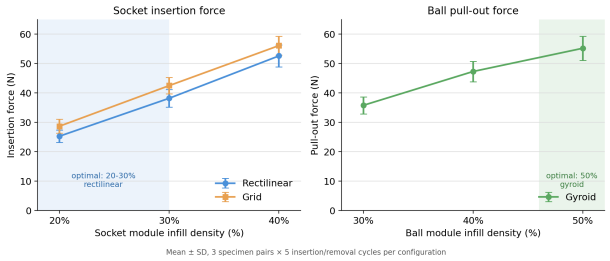


Fig. 2. Force relationships with infill density. Left: Insertion force decreases with lower socket infill. Right: Pull-out force increases with higher ball infill. Shaded region indicates optimal operating range balancing easy insertion with high retention.

B. FEA Validation

FEA provided quantitative validation of failure modes (Table IV). Torsional loading creates shear stress causing progressive disengagement at ∼45° rotation—providing tactile warning before complete separation. Bending simulations confirmed stable connection under forces representative of typical user interactions.

TABLE IV
FEA Simulation Results Summary

Load Case	Applied Load	Key Finding
Torsion	4–5 N·m	Separation at ∼45° rotation
Light Bending	0.50 N	Minimal stress, stable
Heavy Bending	2.54 N	Adequate safety margin
Tension	Variable	Progressive disengagement

C. Negative Results and Lessons Learned

Systematic documentation of failures provided critical design guidance (Table V). The most significant negative result involved TPU tolerance: initial connectors based on CAD specifications did not fit properly. Root cause analysis identified insufficient design tolerances combined with TPU's hyperelasticity not being accurately modeled. The critical lesson: soft materials require 2–3× the design tolerance of rigid parts.

TABLE V
Documented Failure Modes and Design Responses

Failure	Observation	Response
Torsion weakness	Unlocks at 45° rotation	Need rotational constraint; consider magnets
Mfg. variance	±0.3 mm in features	Design for larger tolerances
Combined load	60% strength loss vs. single-axis	Always test multi-modal states

D. Surprising Result: Compliance Enhances Robustness

The most counterintuitive finding was that connector compliance enhances rather than degrades connection robustness through four synergistic mechanisms:

1) Increased contact area: Soft socket walls conform to the ball surface under load, creating a larger contact patch than a rigid socket would allow.
2) Stress distribution: Inherent compliance distributes load across the entire contact surface rather than concentrating at discrete points.
3) Misalignment tolerance: Soft connectors self-align during insertion, tolerating angular misalignment of ±15° compared to only ±3° for rigid snap-fits.
4) Vacuum retention: The soft socket creates a partial vacuum during pull-out as it deforms, adding pneumatic retention force to mechanical interference.

This finding challenges the conventional assumption that connection strength requires connector rigidity.

IV. Conclusions

This work demonstrates that the Connector Compliance Paradox in modular soft robotics can be resolved. MOGO achieves CCR ≈ 1–10 (compared to >100 in existing systems) through uniform TPU construction with a ball-and-socket geometry. Key performance metrics include 55 N pull-out force, 25 N insertion force, force ratio of 2.2, 2-second assembly time, and >500-cycle durability.

The principal insight—that compliance enhances rather than degrades connection robustness—challenges conventional assumptions requiring rigid connectors. The 45° torsional unlock threshold remains a limitation, suggesting future work on keyed ball-and-socket variants or magnetic augmentation for rotational constraint.

Author Contributions

Y. Liu: Ideation, Writing, 3D Modeling/Printing, Manufacturing Process Development. R. Singh: Literature Research, FEA Simulation, Writing, Investigation. H. Liu: Ideation, Conceptualization, 3D Modeling, Writing. B. Stump: Mechanical Ideation, FDM Expertise, 3D Modeling.

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